

Overview

This is the first of a 2-labs sequence working with real-world data from your own data or **particle physics, astronomy, neuroscience**, in part provided by [A3D3](#), an NSF HDR institute. Before starting this lab, you need to first decide which dataset you and your lab partner will work on.

You should have teamed up with at least one other member and agree with the instructors about your dataset.

Lab 8 Instructions

The objectives of this lab are three-fold:

1. Understanding the data,
2. understanding the task, and
3. developing an initial plan for your project.

For a successful Deep Learning project, it's essential to fulfill these objectives before designing/training the neural network model. These objectives will also serve as a systematic roadmap for the project.

Below is the list of questions you want to address for each objective.

1. Understanding the data
 - a. Data size and structure: How big is the data? How is it structured? Are there any obvious patterns? If so, are they expected? Any unused features? Etc.
 - b. Data cleanness Is some of the data corrupted? Is some data mislabeled? If it's simulated, does the simulator have any limitations
 - c. Diversity of features: How many features (e.g., # of pixels, # of columns) are there per data sample? Does your dataset have approximately equal number of datapoints per label?
2. Understanding the task
 - a. What problem are you trying to solve with your neural network model? (e.g., classification or regression?)
 - b. What is the significance of solving the problem?
 - c. What will be the evaluation metric for your neural network model?
 - d. What would constitute a good performance?
 - e. Is there a baseline method to compare against after meeting the performance? (e.g., classical machine learning method, other neural network models)
3. Developing an initial plan for your project.
 - a. What will be the inputs and outputs (including their dimensions) of your neural network model? • What data preparation methods do you plan to use?

- b. What neural network model do you plan to use? (FCN, CNN, RNN, etc)
- c. What loss function do you plan to use?
- d. How do you plan to evaluate your model?

Lab 8 Checkout and Grading

- There will be no written report submissions for the lab8.
- You are expected to work on Lab8_Template.ipynb during the lab session, discuss three Objectives with your TAs, and submit it by the end of your lab session.
- Your grade is based on
 - In-person participation
 - Lightning talks
 - Responses to Three objectives